Name:

ID:

Xi'an Jiaotong-Liverpool University
西交利物浦大学

Module code: CHE113

Module title: Introductory Organic Chemistry

Semester 1, 2024/25, Year 2

Coursework 2

Module Examiner: Dr. Eric Amigues

Department of Chemistry, Tel. 0512-88167123

INSTRUCTIONS to CANDIDATES

1. The total number of marks is 100. The numbers in brackets indicate the marks assigned to the question.

2. Answer **ALL** questions

3. Submit your answers as pictures in ONE pdf file and uploaded to Learning Mall Online with the name CHE113_IDNo_Name_CW2.

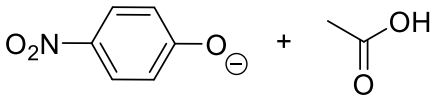
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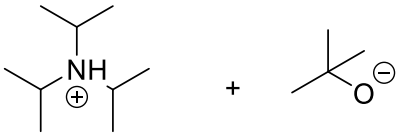
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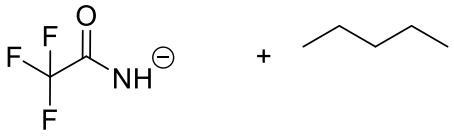
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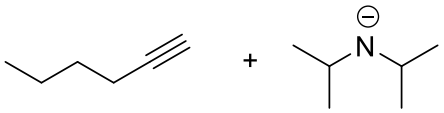
Problem 1 **[30 marks]**

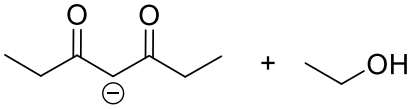
Draw the products formed from the following acid base pairs and use approximate pKa values to predict in which direction the equilibrium will lie. (2 points for each reaction)

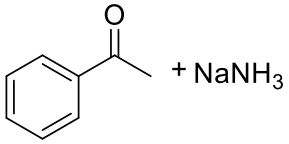
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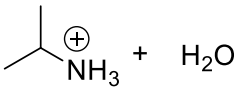
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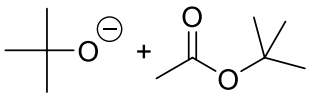
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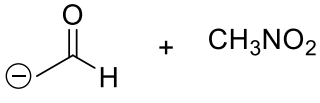
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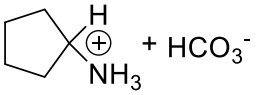
5. 

6. 

7. 

8. 

9. 

10. 

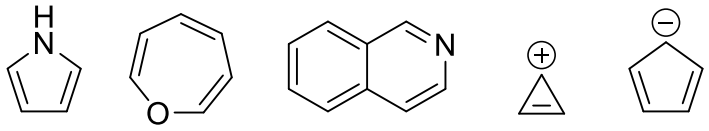
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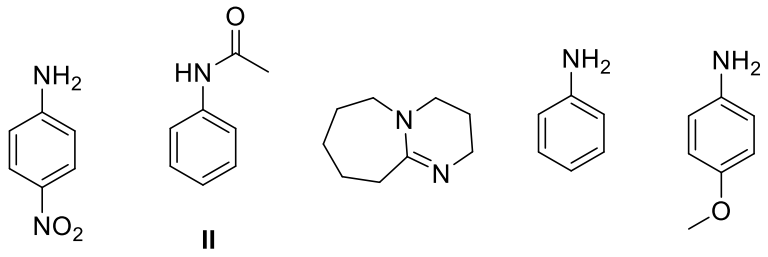
Problem 2 **[10 marks]**

Which of the following compounds are aromatic? Fully justify your answer. (5 compounds, 2 marks each).



Problem 3 **[10 marks]**

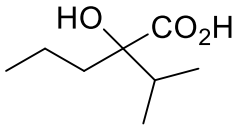
Rank compounds compounds I to V in order of increasing basicity. Explain your answers.



I II III IV V

Problem 4 **[10 marks]**

Propose a multistep synthetic route leading to compound A by using nPrCHO (Butanal) and any other reagents you might need. No mechanisms required



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Problem 5 **[40 marks]**

Provide a mechanism for the following reactions:

